Antibody Effector Mechanisms & Mucosal Immunity

Lecturer: Jason Cyster

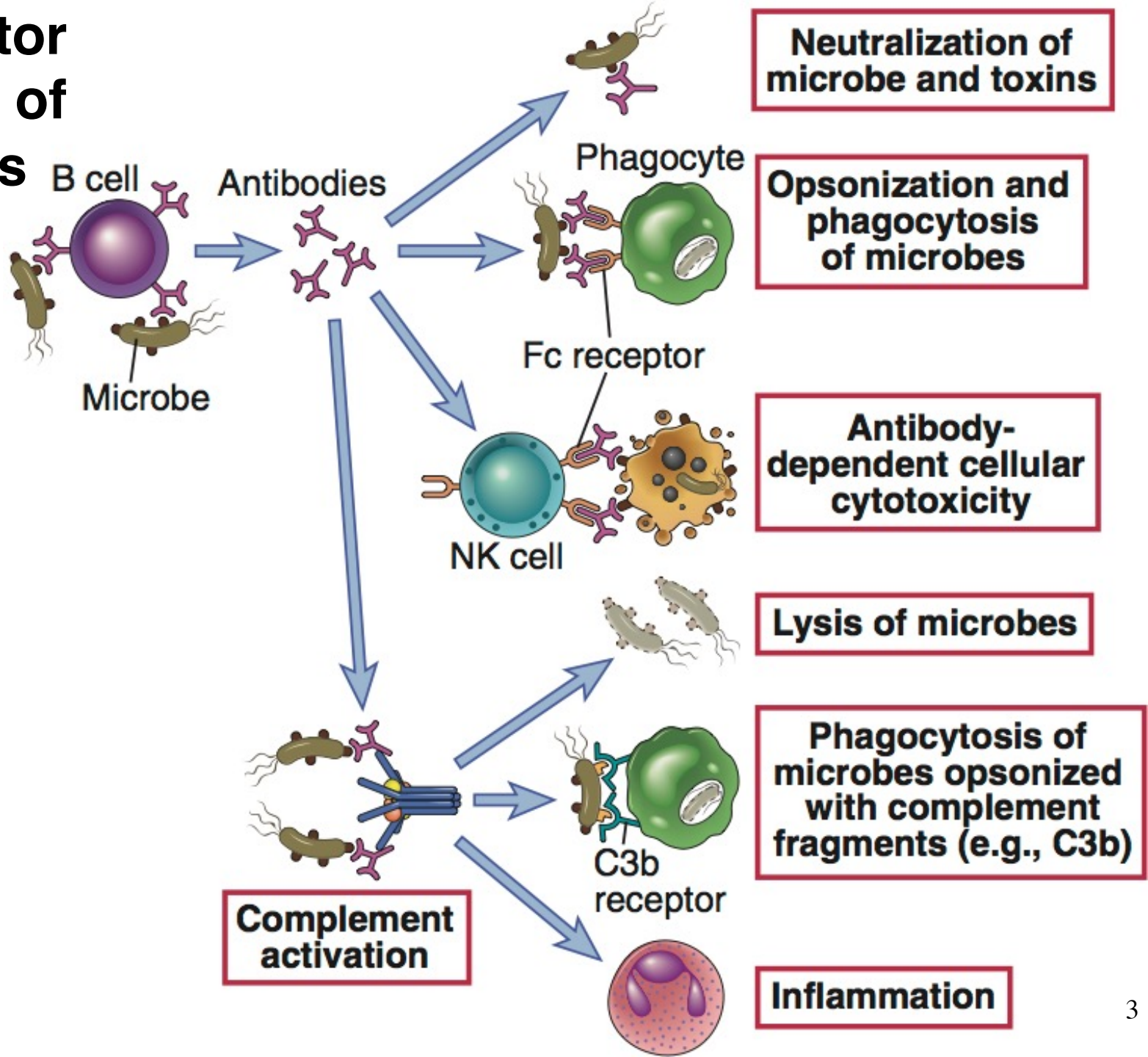
1. Effector mechanisms of antibodies

2. Immune processes at mucosal surfaces

Session Objectives

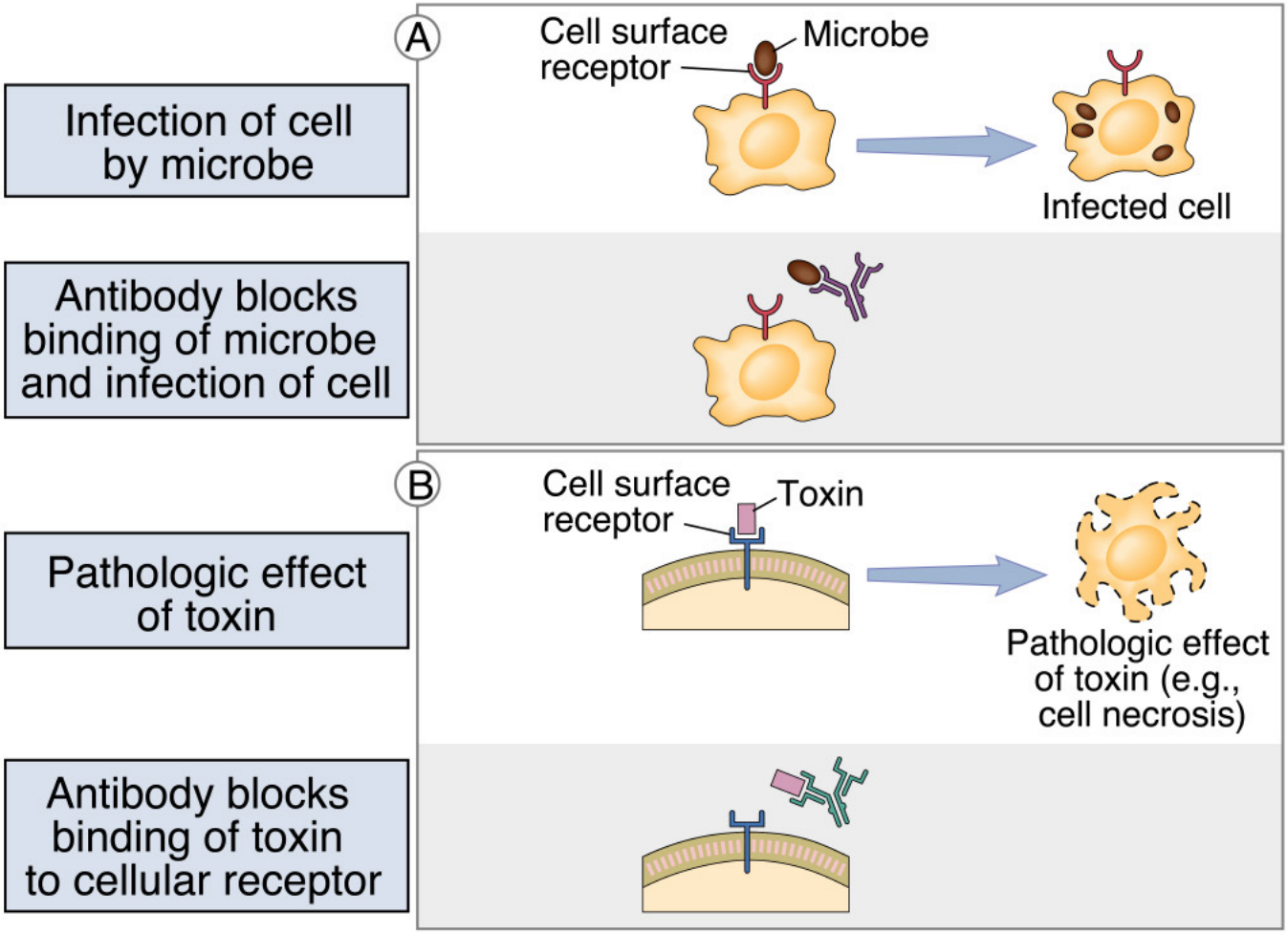
- Describe roles and consequences of antibodies in neutralization, opsonization, antibody-dependent cellular cytotoxicity, and phagocytosis
- Describe how complement is activated and the major actions of the complement system
- Describe how various classes of antibodies can enter secretions and cross epithelial layers or cross into the placenta and why this is important
- Compare and contrast active and passive immunity
- Give examples of molecules, enzymes, and cells that are important for establishing and maintaining mucosal immunity and describe the roles they play

The effector functions of antibodies

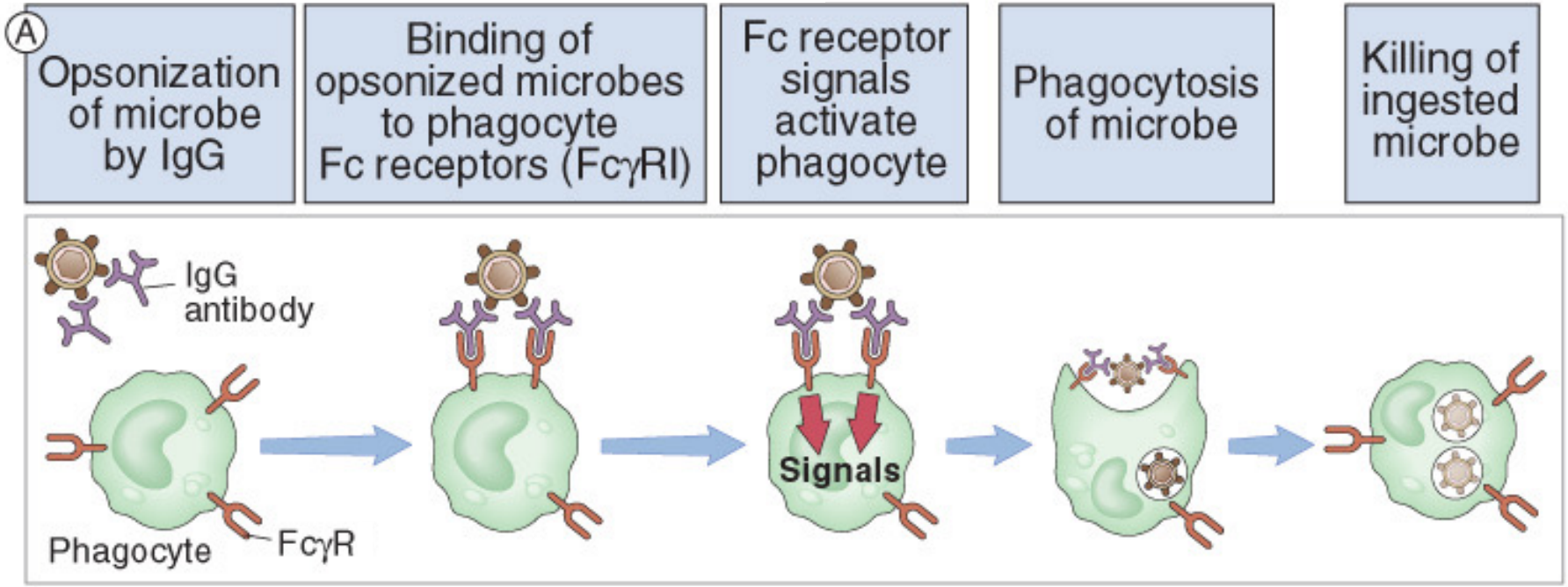


Abbas Fig. 8-1

Effector Functions of Antibodies: Neutralization



Antibody-mediated opsonization and phagocytosis of microbes



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Phagocytic cells (macrophages, neutrophils) express activating Fcγ receptors - when crosslinked they transmit signals that promote engulfment and increased bactericidal activity

The Complement (C') System

- Complement system opsonizes antigens for phagocytosis and can promote direct lysis of some bacteria
- All the C' components pre-exist in an inactive form in the blood (mostly made in the liver); C3 is the most abundant
- Enzymatic (proteolytic) cascade - initial signal strongly amplified
- When a C' component is cleaved, the larger 'b' fragment is chemically labile and becomes covalently bound to nearby surfaces; the smaller 'a' fragment is soluble and can diffuse away
- Can be activated by antigen-bound IgM or IgG (Classical Pathway), by bacterial surfaces (Alternative Pathway) or by mannose-binding lectin (MBL pathway)

Activation of Complement (C')

Classical Pathway

Antibody/Antigen Complex & C1

↓

Various complement components activated to generate

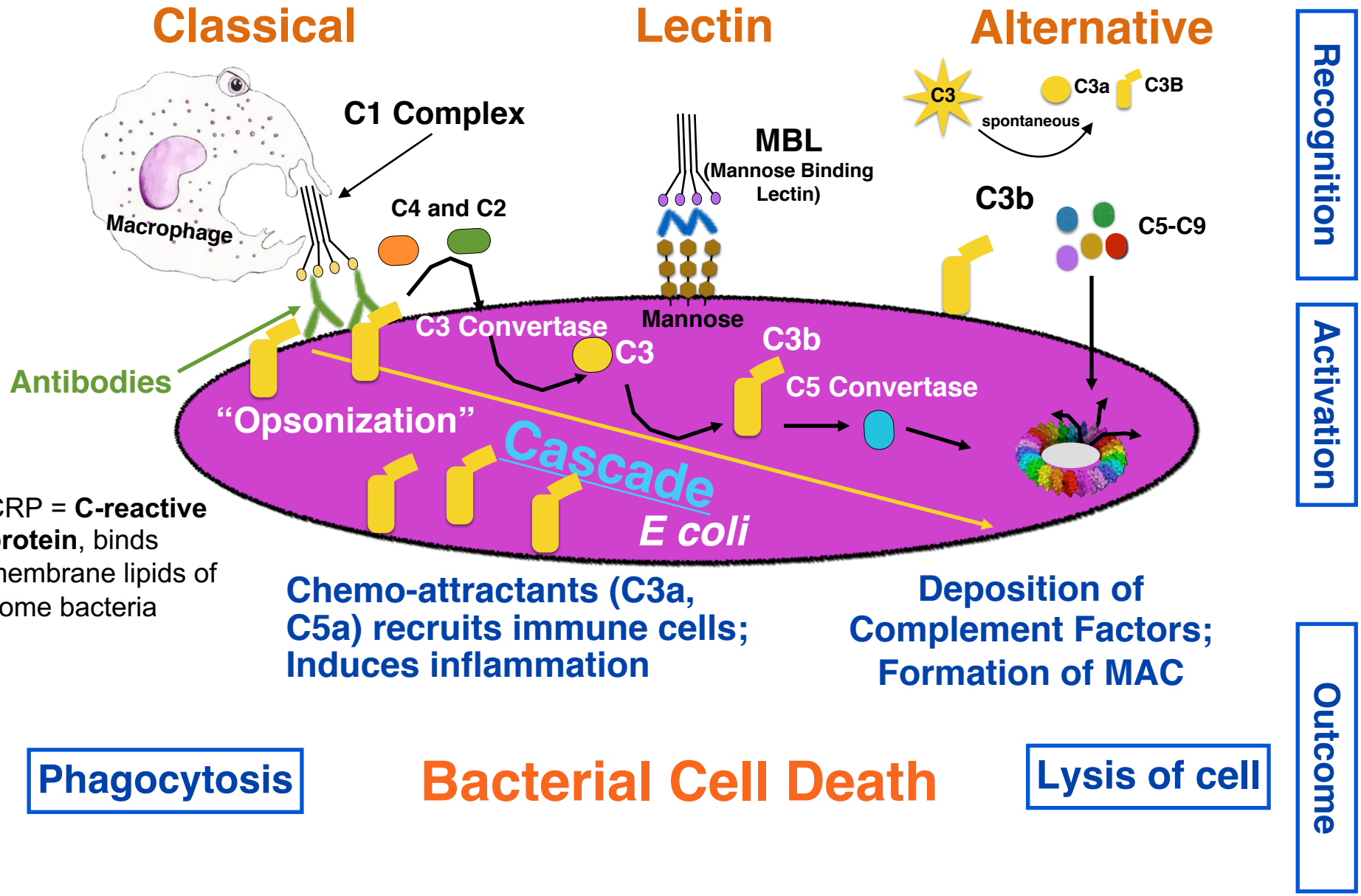
↓

"C3 convertase"

C3 ⟶ C3b + C3a

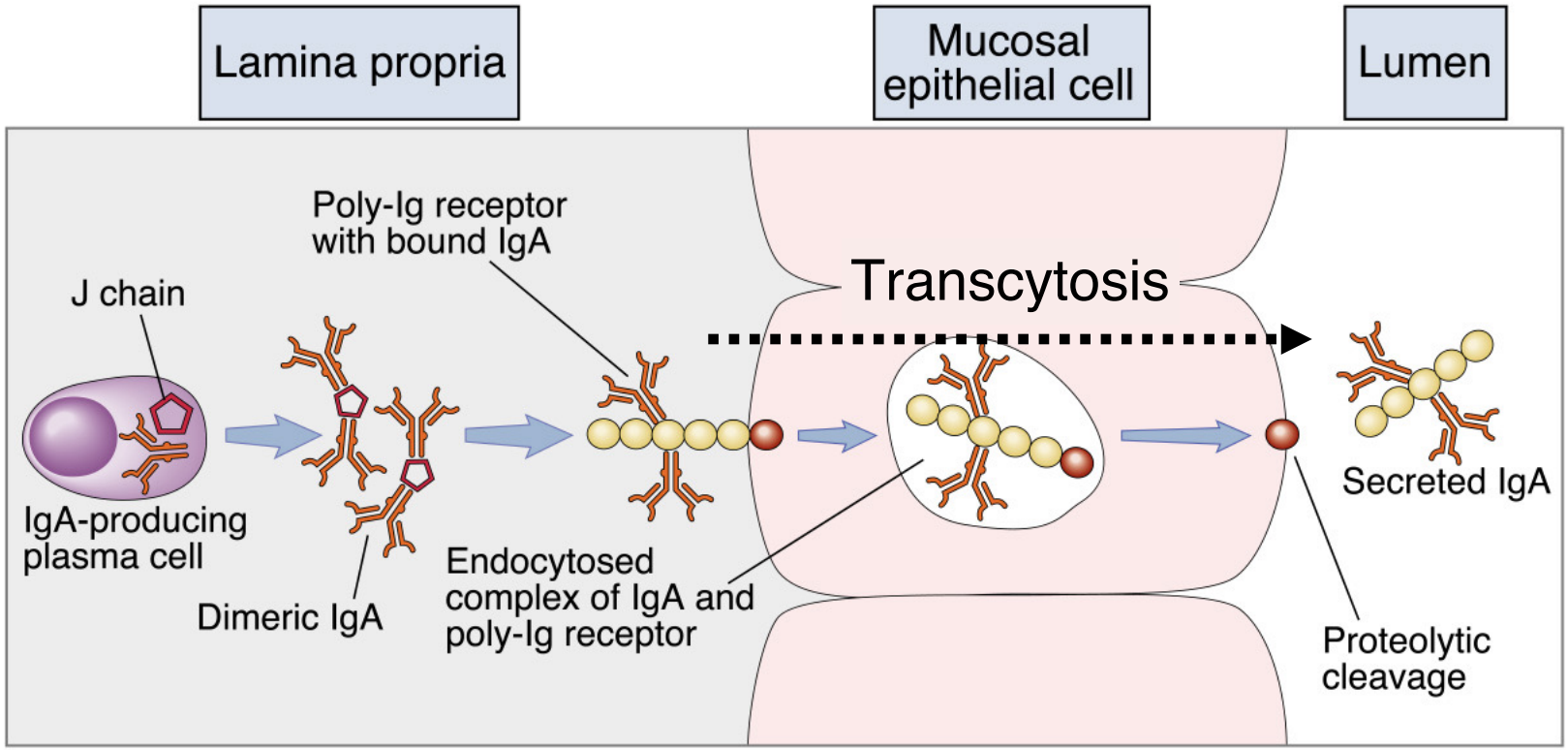
- ➢ C3b decorates surface to promote phagocytic clearance
- ➢ some C3b forms a C5 convertase, generating C5b and C5a
- ➢ C5b causes formation of the Membrane Attack Complex (C5-C9)
- ➢ C3a and C5a chemoattract phagocytic cells

Complement Cascade (Simplified)



Note: the pink color is the surface membrane of the bacterium

Transport IgA through epithelium



- The poly-Ig receptor is a special Fc receptor that binds dimeric IgA
- The IgA released into the gut lumen remains associated with part of the poly-Ig receptor (known as the *secretory component*) and this provides protection against proteolysis by gut proteases
- Secretory IgA functions by neutralization, blocking pathogen binding to the epithelium

Neonatal Immunity

- Maternal IgG is transported across the placenta to the fetus
- Confers passive immunity in the newborn
- The duration of protection is 3-4 months (or ~5 IgG half-lives)
  - explains the high incidence of disease after this period by bacteria such as *Haemophilus influenzae*
- Human milk contains IgA
  - provides some protection against gut pathogens
- Neonatal protection is only as good as the titer of IgG (and IgA) in the mother against the specific organism

If you have a bacterial infection that penetrates into tissue what effector mechanisms of antibodies do you think might help protect?

Complement fixation

Opsonization for Phagocytosis

Neutralization

All of the above

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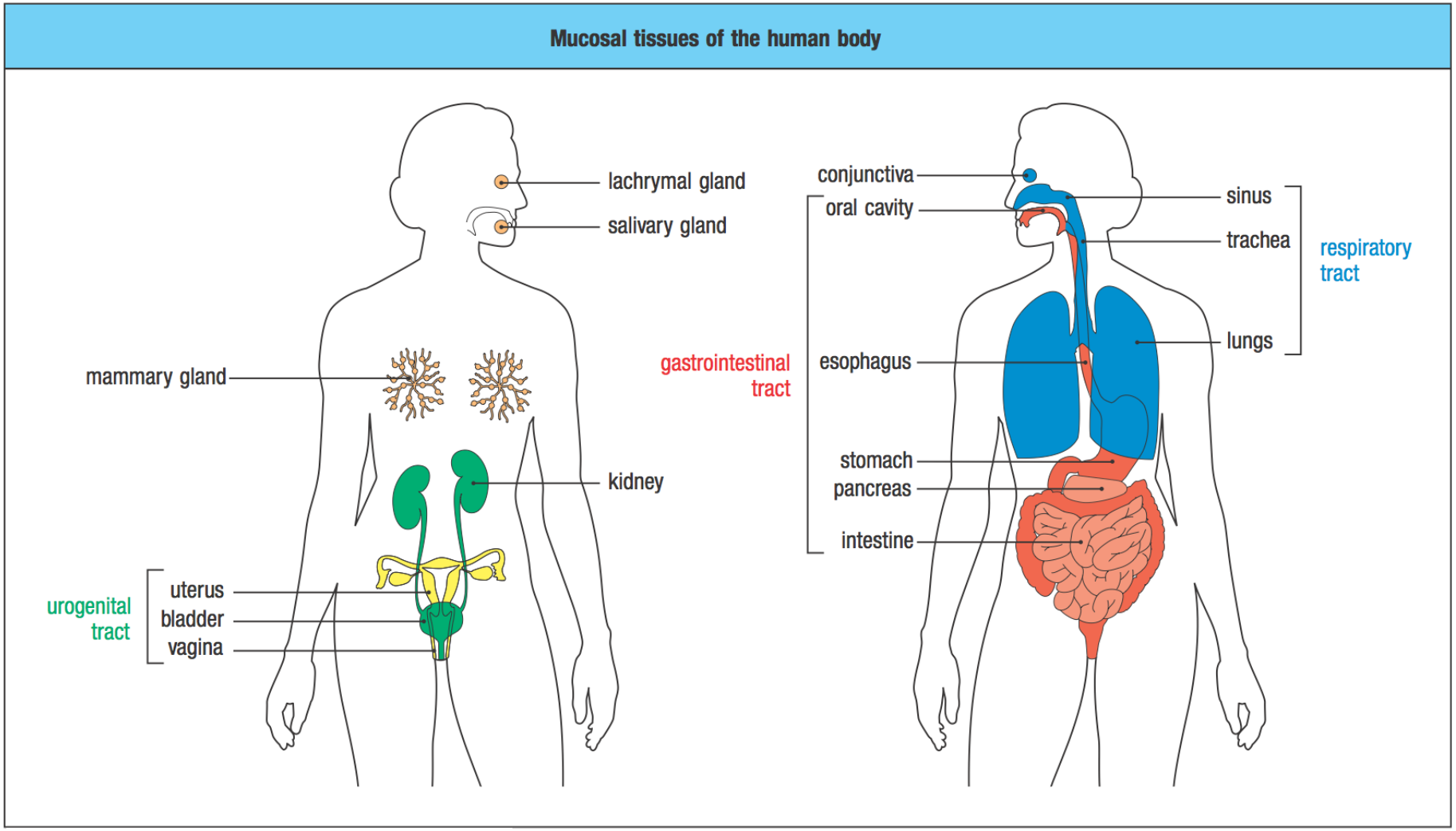
Mucosal tissues – where pathogens most often infect



Parham Figure 10.1, The Immune System, 4th ed.

Special properties of mucosal immunity

Mucus

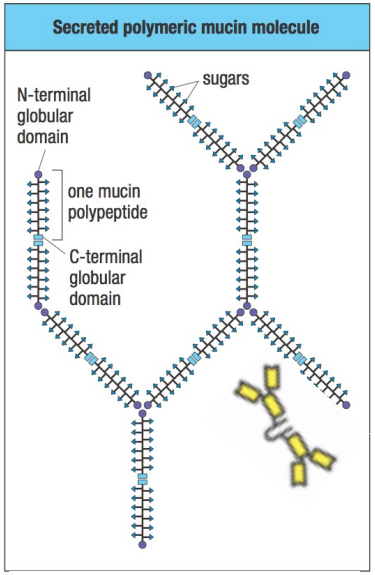
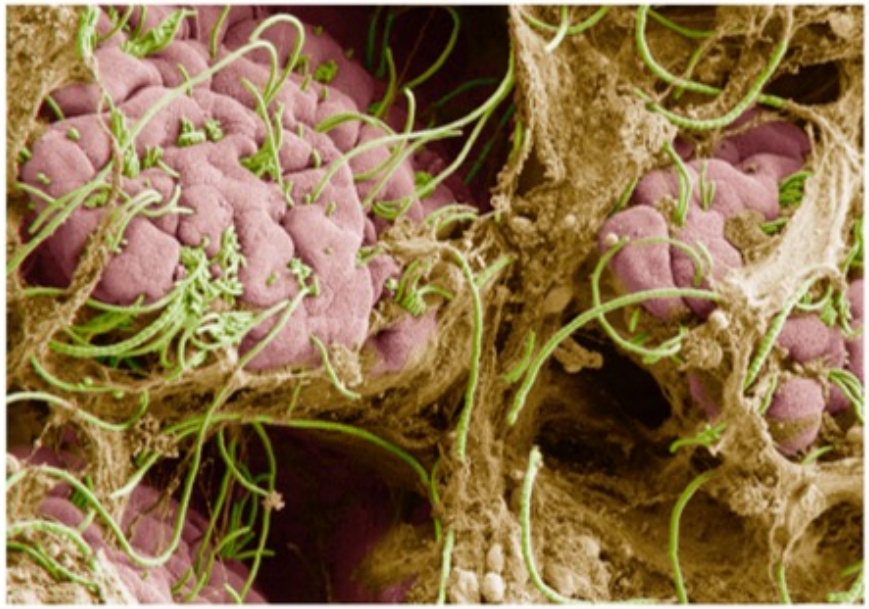


Figure 10.2, The Immune System, 4th ed.

Bacteria in intestine



MucosalImmunology

Microbiota

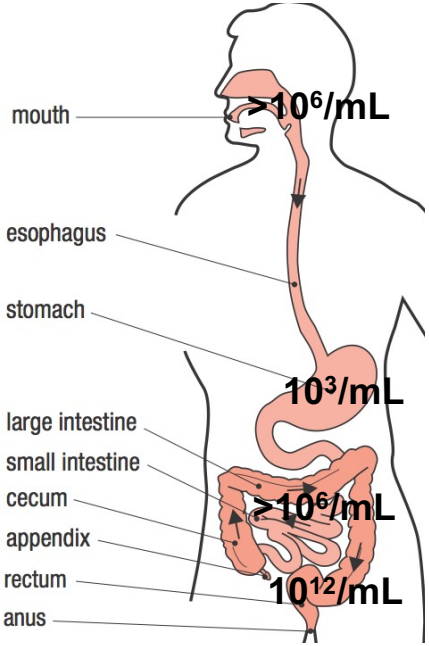
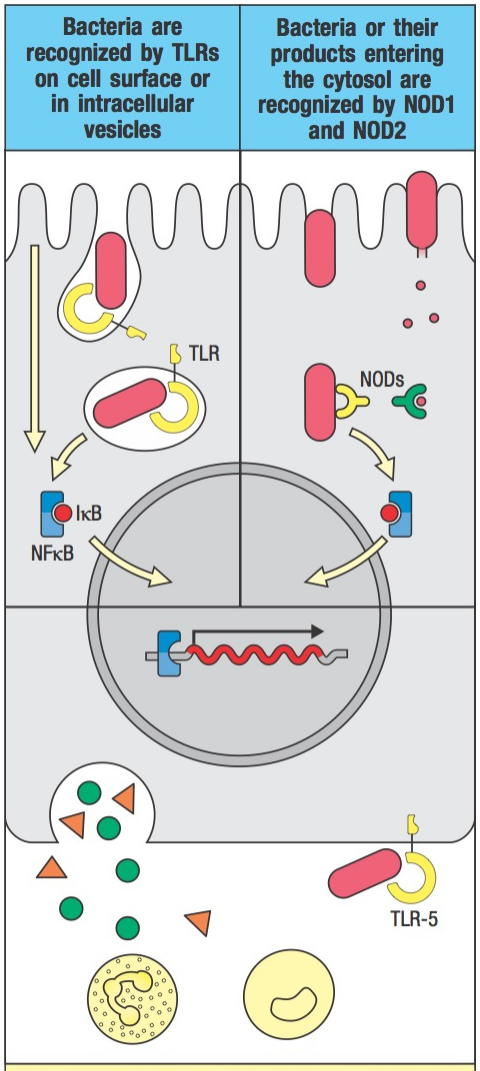


Figure 10.5, The Immune System, 4th ed.

Forms a barrier
"Weep and sweep" removes pathogens, toxins,
binds IgA, defensins

Metabolic functions
Shapes immune system

Special properties of mucosal immunity



TLRs, NOD1, and NOD2 activate NFκB, inducing the epithelial cell to express inflammatory cytokines, chemokines, and other mediators. These recruit and activate neutrophils and monocytes

Figure 10.10, The Immune System, 4th ed.

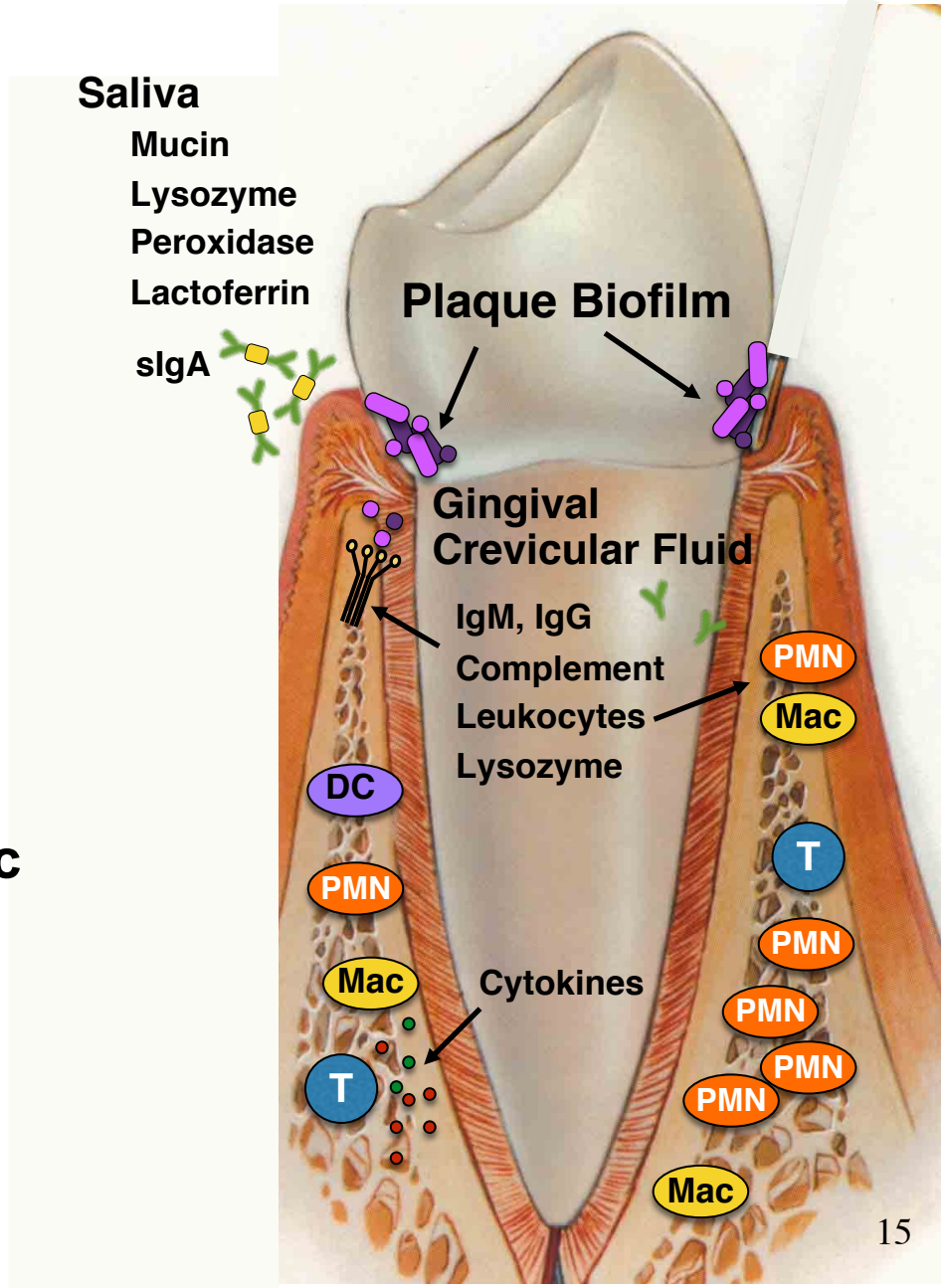
Epithelial cells

- Form a barrier
- Transport antibodies (IgA as well as IgM, IgG)
- Secrete mucins, antimicrobial peptides, etc.

- **Express pattern recognition receptors** including TLRs and NOD1/2 to detect infection
- Bacterial recognition by these receptors increases secretion of anti-microbial peptides and induces immune cell chemoattractants
- Genetic defects in NOD2 strongly predispose to inflammatory bowel disease (Crohn's disease)

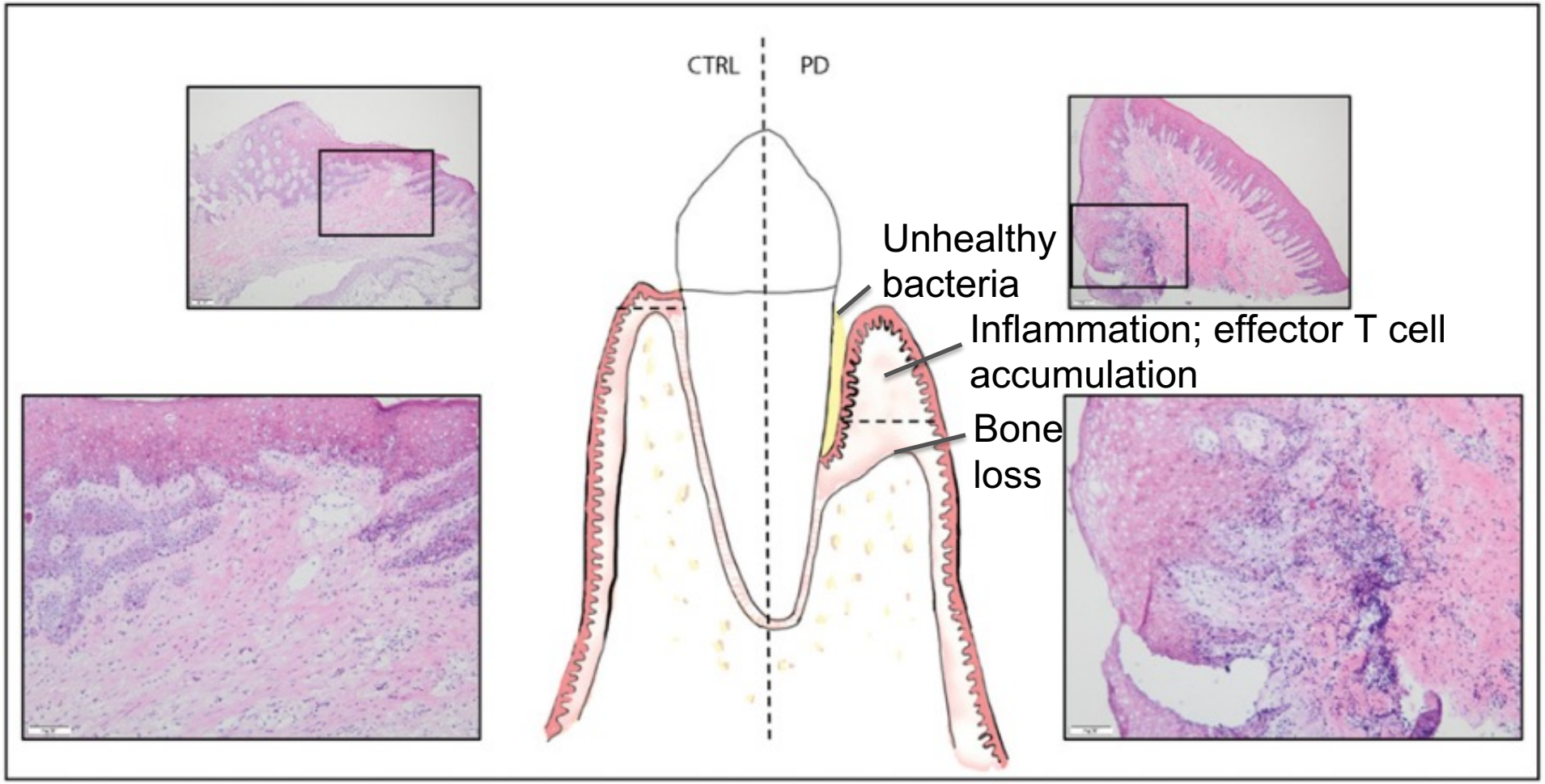
Nonspecific and Specific host defense factors of the mouth

- >10^6/mL bacteria (~200 species)
- **Mucins:** protective, viscous gel covering the oral epithelium.
- **Saliva flow** – we generate and ingest ~1.5 L of saliva per day
- **Defensins, Lysozyme, Proteases, Lactoferrin, Complement**
- **Secretory IgA**
- Tissue **macrophages** and **dendritic cells** beneath the epithelium



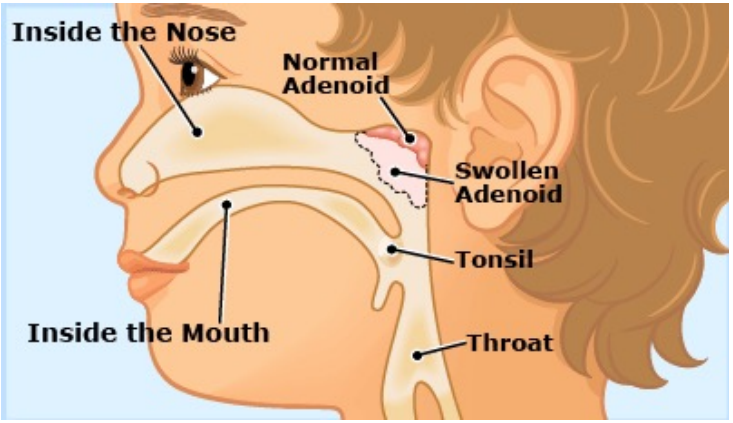
Gingival Tissue Inflammation in Periodontitis

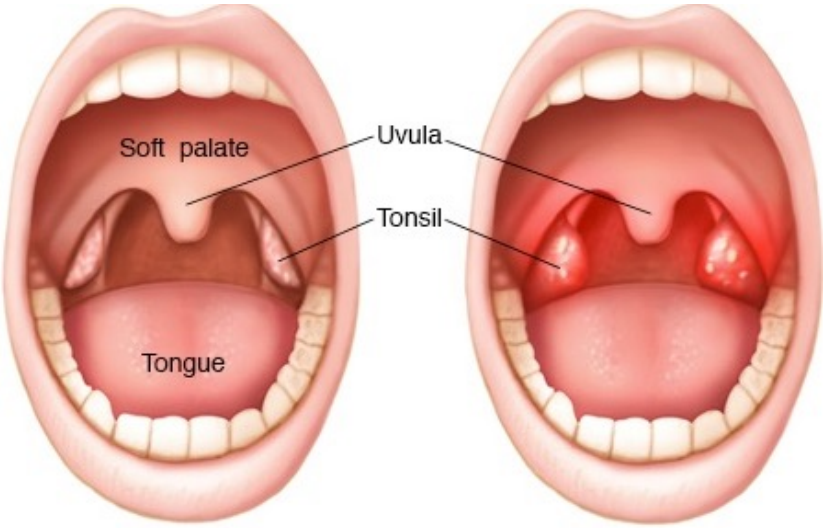
- The destructive, chronic inflammation that manifests clinically as **periodontitis** is caused by an excessive and inappropriate immune response to pathogenic plaque bacteria, coupled with a failure of the normal processes that limit inflammation and drive tissue repair.



- dark blue staining shows activated macrophages producing inflammatory mediators

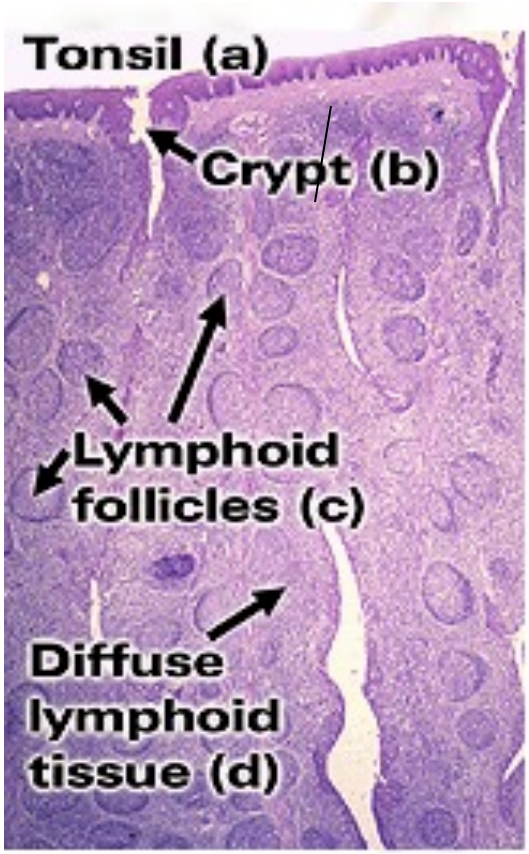
Oral-pharyngeal cavity lymphoid tissues



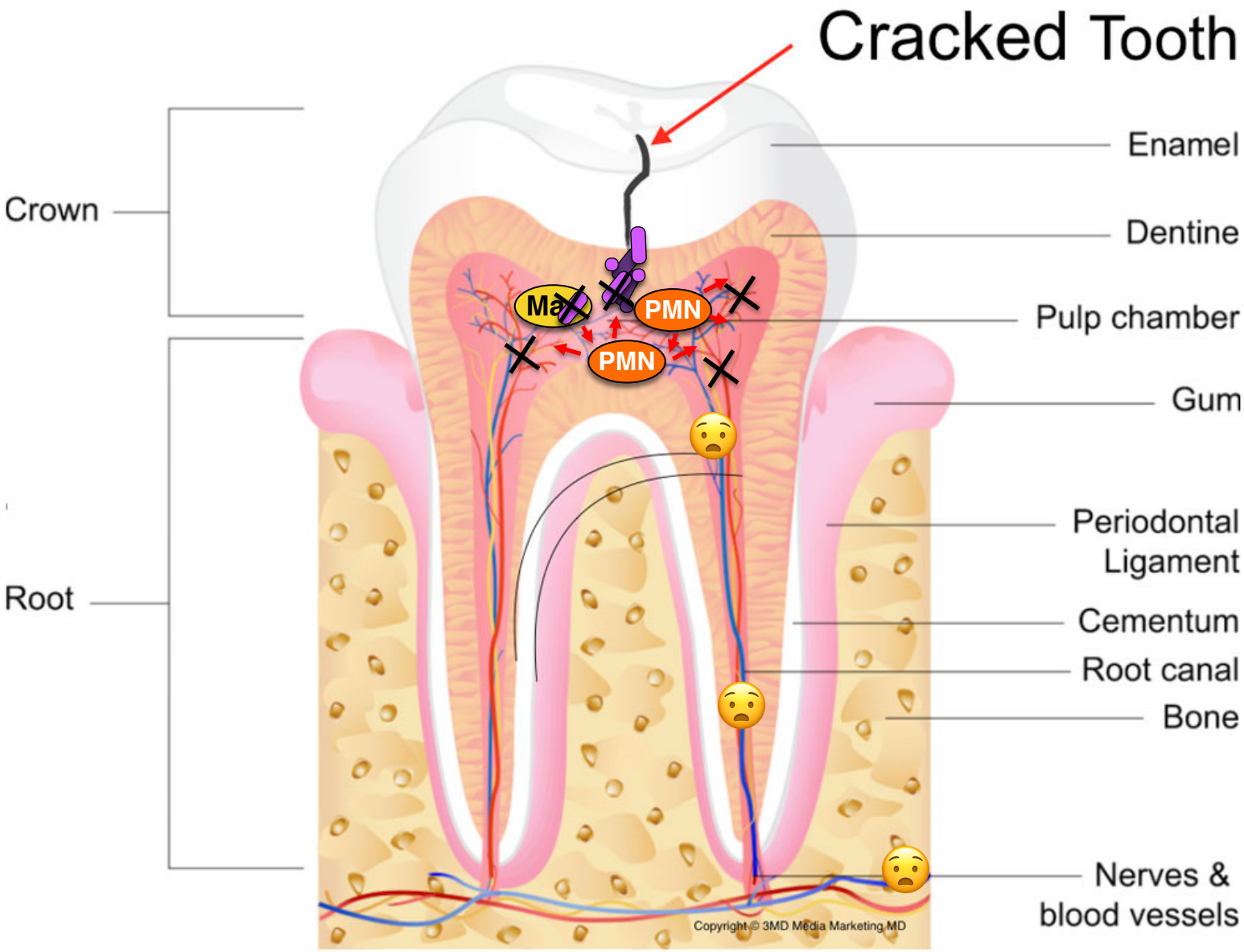


Normal tonsils Inflamed tonsils

squamous epithelium



- The tonsils are lymphoid tissues located towards the back of the mouth immediately beneath the epithelium; they comprise the palatine-, lingual- and tubal-tonsils as well as the adenoids (pharyngeal tonsils)
- During infection tonsils and facial LNs generate IgA producing plasma cells that home to the oral cavity lamina propria and secrete IgA into saliva, etc.



https://www.southlakeendo.com/tooth-pain/cracked-teeth/

During a primary infection of the gingival epithelium, which factors are likely to help control the infection in the first hours?

Neutrophils and Macrophages

B cells and T cells

Complement and defensins

A and B

A and C

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